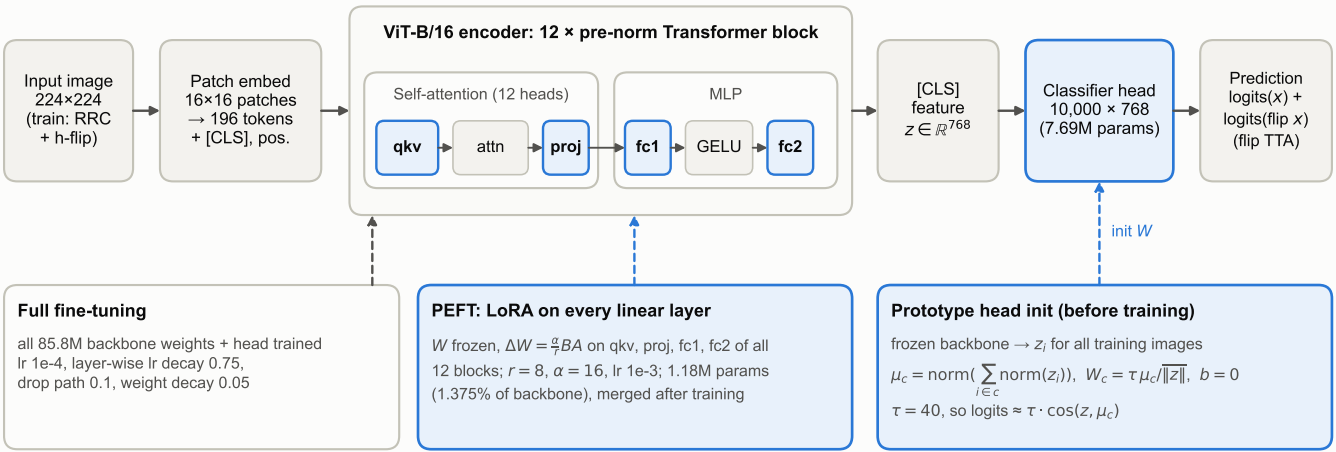

Input image
224×224
(train: RRC
+ h-flip)
Patch embed
16×16 patches
→ 196 tokens
+ [CLS], pos.
ViT-B/16 encoder: 12 × pre-norm Transformer block
Self-attention (12 heads)
qkv
attn
proj
MLP
fc1
GELU
fc2
[CLS]
feature
$z \in \mathbb{R}^{768}$
Classifier head
10,000 × 768
(7.69M params)
Prediction
logits(x) +
logits(flip x)
(flip TTA)
init W
Full fine-tuning
all 85.8M backbone weights + head trained
lr 1e-4, layer-wise lr decay 0.75,
drop path 0.1, weight decay 0.05
PEFT: LoRA on every linear layer
W frozen, $\Delta W = \frac{\alpha}{r} BA$ on qkv, proj, fc1, fc2 of all
12 blocks; $r = 8$, $\alpha = 16$, lr 1e-3; 1.18M params
(1.375% of backbone), merged after training
Prototype head init (before training)
frozen backbone → z_i for all training images
$\mu_c = \mathrm{norm}(\sum_{i \in c} \mathrm{norm}(z_i))$, $W_c = \tau \mu_c / \overline{\|z\|}$, $b = 0$
$\tau = 40$, so logits ≈ $\tau \cdot \cos(z, \mu_c)$